Math 56 Fall 2024
Otis Chodosh

# PROJECT 2 (REAL ANALYSIS)

ELSA ZHOU

## 1. Introduction

In this paper, we seek to explore the effect of the rearrangement of an infinite series on its sum. We will begin by defining a rearrangement. Then, we will go on to show that for a series with nonnegative terms, the infinite sum equals the supremum of the sums over all finite subsets of $\mathbb{N}$. As an important corollary, we will show that any rearrangement of this series with nonnegative terms converges to the same limit.
We will then go on to look at series with mixed signs. Specifically, we will define the sums of the positive and negative parts of an infinite series, and show how absolute convergence relates to those sums. Importantly, we will show that if either sum is finite, any rearrangement of the series converges to the same limit. This is a key takeaway.
We will then provide some examples of rearrangements, including one that we rearrange to converge to a different limit than the original series.

## 2. Definition of Rearrangement

**Definition 2.1.** Let $(a_n)$ be a sequence. An *infinite series* is a formal expression of the form

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots.$$

We define the corresponding sequence of partial sums $(s_m)$ by

$$s_m = a_1 + a_2 + a_3 + \cdots + a_m,$$

and say that the series $\sum_{n=1}^{\infty} a_n$ converges to $a$ if the sequence $(s_m)$ converges to $a$. In this case we write $\sum_{n=1}^{\infty} a_n = a$.

A rearrangement of a series is determined by using the same terms, but changing the order in which these terms occur.

**Definition 2.2.** This definition is directly from Abbott 2.7.9: Let $\sum_{n=1}^{\infty} a_n$ be a series. A series $\sum_{n=1}^{\infty} b_n$ is called a rearrangement of $\sum_{n=1}^{\infty} a_n$ if there exists a one-to-one, onto function $f : \mathbb{N} \to \mathbb{N}$ such that $b_{f(n)} = a_n$ for all $n \in \mathbb{N}$.

**Example 2.3.** This section follows the outline. Consider the alternating harmonic series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots.$$

One rearrangement that we could apply to this series is where, instead of alternating one positive term and one negative term, we could create a rearrangement where we alternate two positive terms and one negative term. In other words,

$$\sum_{n=1}^{\infty} b_n = 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots.$$

We want our terms in $b_n$ to correspond with our terms in $a_n$. Let us now create a function $f : \mathbb{N} \to \mathbb{N}$ defined in the following way:
Let $k \in \mathbb{N}$. Let the $k$-th positive term in $b_n$ equal $f(2k-1) = a_{(2k-1)}$. For example, the 2nd positive term in $b_n$ is equal to $a_{2(2)-1} = a_3$. Indeed, we see that the second positive term in $b_n$ is $b_2$, and $b_2 = \frac{1}{3} = a_3$.
Let the $k$-th negative term in $b_n$ equal $f(2k) = a_{2k}$. For example, the 2nd negative term in $b_n$ is equal to $a_{2(2)} = a_4$. Indeed, we see that the second negative term in $b_n$ is $b_6$, and $b_6 = \frac{-1}{4} = a_4$.

We now want to show that our function is bijective. Intuitively and geometrically, we might see that the two functions that describe our positive and negative terms respectively are linear functions with range $= (-\infty, \infty)$, so we see that $f$ is bijective. However, more exactly, let us now try to prove that $f$ is bijective.

Injectivity: $f(2k-1)$ describes our $k$-th positive term of $b_n$, and $f(2k)$ describes our $k$-th negative term of $b_n$, where $f(c) = a_c$. As $f(2k-1)$ produces only odd numbers, it will index all of $a_n$ with an odd index (i.e. $a_1, a_3, a_5, \dots$). As $f(2k)$ produces only even numbers, it will index all of $a_n$ with an even index (i.e. $a_2, a_4, a_6, \dots$). Thus, we see that if $f(x) = f(y)$, $x$ and $y$ must either be both odd, or both even. If they are both odd, we clearly see that $x = 2k - 1 = y$ has only one solution for $k$, so $x = y$. If they are both even, we clearly see that $x = 2k = y$ has only one solution for $k$, so again $x = y$. Thus, we have injectivity for $f$.

Surjectivity: Every $n \in \mathbb{N}$ is either odd or even. Thus, it can be expressed as either $2k$ or $2k-1$, where $k \in \mathbb{N}$. Thus, every term in $a_n$ is in $b_n$, and vice versa. This makes $f$ surjective.

Together, we see that $f$ is bijective. Specifically, when we say $f$, we mean that $b_n = a_{f(n)}$. This implies that there exists a bijective inverse such that $a_n = b_{f(n)}$. Thus, we have shown that there exists a bijective function $f : \mathbb{N} \to \mathbb{N}$ such that $a_n = b_{f(n)}$ for all $n \in \mathbb{N}$.

## 3. Rearrangement of series with non-negative terms

**Proposition 3.1.** *If all the terms $a_n \geq 0$, then*

$$\sum_{n=1}^{\infty} a_n = \sup \left\{ \sum_{n \in K} a_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N} \right\}$$

*where the supremum is taken over all finite subsets of $\mathbb{N} = \{1, 2, 3, \dots\}$.*

*Proof.* This section closely follows the Homework 8 Solutions from Canvas.

We set $A = \sup \left\{ \sum_{n \in K} a_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N} \right\}$. We wish to show that $\sup A = \sum_{n=1}^{\infty} a_n$. We have that $a_n \geq 0$, so for any finite subset of $\mathbb{N}$ we have that $\sum_{n \in K} a_i \leq \sum_{n=1}^{\infty} a_n$. Since this is true for any finite subset $K$, we have that $\sum_{n=1}^{\infty} a_n$ is an upper bound for the set $A$. We will show that it is the supremum.

We take $\epsilon > 0$. We can now split into two cases.

**Case 1:** $\sum_{n=1}^{\infty} a_n < \infty$
If the sum is finite, this implies that partial sums converge to a limit. Let $\sum_{n=1}^{\infty} a_n = a$, where $a \in \mathbb{R}$. Thus, the partial sums $s_N = \sum_{n=1}^{N} a_n \to a = \sum_{n=1}^{\infty} a_n$ as $N \to \infty$. This implies that there exists $N_1 \in \mathbb{N}$ such that $|\sum_{n=1}^{N_1} a_n - a| < \epsilon$. Extracting $a$, we have that $a - \sum_{n=1}^{N_1} a_n < \epsilon$. This implies that $a - \epsilon < \sum_{n=1}^{N_1} a_n$.
If we set $K = 1, \dots, N_1$, which would be a finite subset of $\mathbb{N}$, we have that $\sum_{n=1}^{N_1} a_n = \sum_{n \in K} a_n$. This implies that we have found $x = \sum_{n=1}^{N_1} a_n \in A$ such that $a - \epsilon < x$, which implies $a = \sum_{n=1}^{\infty} a_n$ is the supremum of $A$.
**Case 2:** $\sum_{n=1}^{\infty} a_n = \infty$
Let us take $M > 0$. If the sum is infinite, this implies that partial sums diverge to infinity. Thus, the partial sums $s_N = \sum_{n=1}^{N} a_n \to \infty$ as $N \to \infty$. This implies there exists an $N_2 \in \mathbb{N}$ such that $\sum_{n=1}^{N_2} a_n > M$. If we set $K = 1, \dots, N_2$, which would be a finite subset of $\mathbb{N}$, we have that $\sum_{n=1}^{N_2} a_n = \sum_{n \in K} a_n$. This implies we have found $x = \sum_{n=1}^{N_2} a_n \in A$ such that $x > M$. Since $M$ was arbitrary, we have that $\sup A = \infty = \sum_{n=1}^{\infty} a_n$.

Thus, in all cases, $\sum_{n=1}^{\infty} a_n = \sup \left\{ \sum_{n \in K} a_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N} \right\}$.

□

**Corollary 3.2.** *Any rearrangement of this series converges to the same (possibly infinite) limit.*

*Proof.* This section follows the outline. As defined in Proposition 3.1, we have that

$$\sum_{n=1}^{\infty} a_n = \sup\left\{\sum_{n\in K} a_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N}\right\}$$

Let $b_n$ where $n \in \mathbb{N}$ be the rearrangement. By definition, there exists a bijective function $f : \mathbb{N} \to \mathbb{N}$ such that $b_{f(n)} = a_n$. Since a rearrangement would not change the sign of any terms, all the terms in $b_n \geq 0$. Thus, Proposition 3.1 also applies to $b_n$, and we know that

$$\sum_{n=1}^{\infty} b_n = \sup\left\{\sum_{n\in K} b_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N}\right\}$$

Let $K$ be a finite subset of $\mathbb{N}$. Because $b_{f(n)} = a_n$ is bijective, we know that every element in the image of $f$, or $b_n$, is also in the pre-image of $f$, or $a_n$. This implies the sum $b_n$ over $K$ is equal to the sum $a_n$ over $f^{-}1(K)$. (We know that $f^{-}1(K)$ is a finite subset of $\mathbb{N}$ because $f$ is bijective.) In other words,

$$\sum_{n\in K} b_n = \sum_{n\in K} a_{f^{-}1(n)}$$

Similarly, if $J$ is a finite subset of $\mathbb{N}$, we can use $b_{f(n)} = a_n$ to say that

$$\sum_{n\in J} a_n = \sum_{n\in J} b_{f(n)}$$

We see that summing $a_n$ over a finite subset $J$ is equal to summing $b_n$ over $f(J)$. Similarly, because $f$ is bijective, we are guaranteed that $f(J)$ is also a finite subset of $\mathbb{N}$.

Thus, we see that as a result of $f$ being bijective, the sum over all finite subsets of $b_n$ is one-to-one with the sum over all finite subsets of $a_n$. Thus, we know $K \leftrightarrow f^{-}1(K)$ and $J \leftrightarrow f(J)$ map each other bijectively, so the following sets of sums are the same.

$$\left\{\sum_{n\in K} b_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N}\right\}$$

$$\left\{\sum_{n\in J} a_n \;\middle|\; J \text{ is a finite subset of } \mathbb{N}\right\}$$

Thus, their supremum must also be equal.

$$\sup\left\{\sum_{n\in K} b_n \;\middle|\; K \text{ is a finite subset of } \mathbb{N}\right\} = \sup\left\{\sum_{n\in J} a_n \;\middle|\; J \text{ is a finite subset of } \mathbb{N}\right\}$$

Thus, as defined in Proposition 3.1, we have that

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n$$

□

## 4. Rearrangement of series with mixed signs

**Definition 4.1.** Given a real number $a$, define its *positive part* by $a^+ = \max(a, 0)$ and define its *negative part* by $a^- = \max(-a, 0)$.

**Lemma 4.2.** $a = a^+ - a^-$.

*Proof.* This section follows the outline. Proof by cases:

**Case 1:** $a > 0$
By definition, $a^+ = \max(a, 0)$. Since $a > 0$, we have that $a^+ = a$. By definition, $a^- = \max(-a, 0)$. Since $a > 0$, $-a < 0$, which means that $a^- = 0$. Adding $a^+$ and $a^-$, we arrive at $a - 0 = a$.

**Case 2:** $a < 0$
By definition, $a^+ = \max(a, 0)$. Since $a < 0$, we have that $a^+ = 0$. By definition, $a^- = \max(-a, 0)$. Since $a < 0$, $-a > 0$, which means that $a^- = -a$. Adding $a^+$ and $a^-$, we arrive at $0 - (-a) = a$.

**Case 3:** $a = 0$
By definition, $a^+ = \max(a, 0)$. Since $a = 0$, we have that $a^+ = 0$. By definition, $a^- = \max(-a, 0)$. Since $a = 0$, which means that $a^- = 0$. Adding $a^+$ and $a^-$, we arrive at $0 - 0 = 0 = a$.

Thus, we see in all cases that $a = a^+ - a^-$. □

**Definition 4.3.** Given a series $\sum_{n=1}^{\infty} a_n$ define $S^+ = \sum_{n=1}^{\infty} a_n^+$ and $S^- = \sum_{n=1}^{\infty} a_n^-$.

**Theorem 4.4.** *If $S^+$ and $S^-$ are both finite, then $\sum_{n=1}^{\infty} a_n = S^+ - S^-$.*

*Proof.* This section follows the outline. Given that $S^+$ and $S^-$ are both finite, this means that the partial sums $s_k^+ = \sum_{n=1}^{\infty} a_n^+$ and $s_k^- = \sum_{n=1}^{\infty} a_n^-$ are finite. In other words, these partial sums are bounded as $k$ approaches $\infty$, which implies that the series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converge.

Let us now consider $s_k = \sum_{n=1}^{k} a_n$. We know from Lemma 4.2 that $a_n = a_n^+ - a_n^-$. Thus, since we know that the series $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converge, we can rewrite $s_k$ as the following:

$$\begin{aligned}
\sum_{n=1}^{k} a_n &= \sum_{n=1}^{k} (a_n^+ - a_n^-) \\
&= \sum_{n=1}^{k} a_n^+ - \sum_{n=1}^{k} a_n^- \\
&= s_k^+ - s_k^-
\end{aligned}$$

We now use that $S^+ = \lim_{k\to\infty} s_k+$ and $S^- = \lim_{k\to\infty} s_k-$. Thus, we see that $\lim_{k\to\infty} s_k = S^+ - S^-$. In other words, as $k \to \infty$, the series $\sum_{n=1}^{\infty} a_n$ converges to $S^+ - S^-$. $\square$

**Definition 4.5.** A series $\sum_{n=1}^{\infty} a_n$ is said to be *absolutely convergent* if $\sum_{n=1}^{\infty} |a_n|$ is finite.

**Proposition 4.6.** *$\sum_{n=1}^{\infty} a_n$ is absolutely convergent if and only if $S^+$ and $S^-$ are both finite.*

*Proof.* This section follows the outline.
( $\Longrightarrow$ ) We need to show that if $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then $S^+$ and $S^-$ are both finite.

Let us assume $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. By definition, this means $\sum_{n=1}^{\infty} |a_n| < \infty$. As we proved above in Lemma 4.2, we also define $a_n = a_n^+ - a_n^-$. Thus, we know $|a_n| = a_n^+ + a_n^-$.

This implies that the expression $\sum_{n=1}^{\infty} |a_n|$ can be written as a sum of two terms, as such:

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} (a_n^+ + a_n^-)$$

By definition, we know that $a_n^+ = \max(a_n, 0)$. This implies that $a_n^+ \geq 0$, since if $a_n < 0$, $a_n^+$ returns 0. By definition, we know that $a_n^- = \max(-a_n, 0)$. This similarly implies that $a_n^- \geq 0$, since if $a_n > 0$, $a_n^-$ returns 0. Thus, we know that both $a_n^+$ and $a_n^-$ have only nonnegative terms.

Now, let us return to our expression of sums. In order to split the left hand side into two sums, we first have to show that $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ both converge. We can do this by noting that $\sum_{n=1}^{\infty} (a_n^+ + a_n^-)$ converges, since we know $\sum_{n=1}^{\infty} |a_n| < \infty$. Here, we use a comparison test. We know that both $a_n^+$ and $a_n^-$ are nonnegative. Therefore, when we observe that $a_n^+ < (a_n^+ + a_n^-)$, what follows is also that $\sum_{n=1}^{\infty} a_n^+ < \sum_{n=1}^{\infty} (a_n^+ + a_n^-)$. As a result, we know that $\sum_{n=1}^{\infty} a_n^+$ must converge since $\sum_{n=1}^{\infty} (a_n^+ + a_n^-)$ converges.

Similarly, we can use a comparison test for $a_n^-$. We observe that $a_n^- < (a_n^+ + a_n^-)$, which implies that $\sum_{n=1}^{\infty} a_n^- < \sum_{n=1}^{\infty} (a_n^+ + a_n^-)$, which in turn implies that $\sum_{n=1}^{\infty} a_n^-$ converges. Thus, since we know that both $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ converge, we can rewrite our expression as the following:

$$\begin{aligned} \sum_{n=1}^{\infty} |a_n| &= \sum_{n=1}^{\infty} a_n^+ + \sum_{n=1}^{\infty} a_n^- \\ &= S^+ + S^- \end{aligned}$$

Besides being a useful expression, this also confirms that our two nonnegative terms must be finite, since they add to a finite sum. Specifically, we have shown if $\sum_{n=1}^{\infty} a_n$ is absolutely

convergent, then $S^+$ and $S^-$ are both finite.

( $\Longleftarrow$ ) Now, we want to show that if $S^+$ and $S^-$ are both finite, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

Let us assume that $S^+$ and $S^-$ are finite. As we stated in the first direction, we know $|a_n| = a_n^+ + a_n^-$. Again, as we stated in the first direction, we know both $a_n^+$ and $a_n^-$ to be nonnegative terms. Here, since we are given that $S^+$ and $S^-$ are both finite, we can use the Algebraic Limit Theorem to write the sum $\sum_{n=1}^{\infty} |a_n|$ as the sum of two terms as such:

$$\begin{aligned}\sum_{n=1}^{\infty} |a_n| &= \sum_{n=1}^{\infty} (a_n^+ + a_n^-) \\ &= \sum_{n=1}^{\infty} a_n^+ + \sum_{n=1}^{\infty} a_n^- \\ &= S^+ + S^-\end{aligned}$$

As stated above, we have assumed that $S^+$ and $S^-$ are both finite. Since we have $\sum_{n=1}^{\infty} |a_n|$ as the sum of these two finite terms, this implies that $\sum_{n=1}^{\infty} |a_n|$ is also finite. A very brief proof of that is as follows:

> Let $x$ and $y$ be two nonnegative finite terms, meaning $x < \infty$ and $y < \infty$. Their sum $x + y < \infty + \infty = \infty$. Thus, the sum of two finite terms is also finite.

Thus, we have shown that the sum of these two nonnegative finite terms is also finite. Specifically, we have shown that if $S^+$ and $S^-$ are both finite, $\sum_{n=1}^{\infty} |a_n|$ is also finite. By Definition 4.5, we know that a series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent if $\sum_{n=1}^{\infty} |a_n|$ is finite. Since we have shown that $\sum_{n=1}^{\infty} |a_n|$ is indeed finite here, this implies that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

Therefore, we see that if $S^+$ and $S^-$ are both finite, $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. We have thus proven both directions of the proof, and conclude that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent if and only if $S^+$ and $S^-$ are both finite. $\square$

Note, we also have that that if $S^+$ is infinite while $S^-$ is finite, then $\sum_{n=1}^{\infty} a_n = \infty$; if $S^+$ is finite while $S^-$ is infinite, then $\sum_{n=1}^{\infty} a_n = -\infty$.

**Corollary 4.7.** *Let $\sum_{n=1}^{\infty} a_n$ be a series in which $S^+$ or $S^-$ is finite. Then, any rearrangement of this series converges to the same (possibly infinite) limit.*

*Proof.* This section follows the outline. Let us look at three possible cases to prove this corollary. Let us call the rearrangement of the series $b_k$, which we can express similarly to

$a_n$ as $b_k = b_k^+ - b_k^-$, and where we define $b_k^+$ as the positive terms of the rearrangement and $b_k^-$ as the negative terms of the rearrangement.

**Case 1: $S^+$ is finite, $S^-$ is finite.**

We use Proposition 4.6 here, which tells us that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent iff $S^+$ and $S^-$ are finite. So, since we are given that $S^+$ and $S^-$ are finite, we know that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

We use Theorem 4.4 to write the following:

$$\sum_{n=1}^{\infty} a_n = S^+ - S^-$$

According to Corollary 3.2, we know that for a series $\sum_{n=1}^{\infty} a_n$ with nonnegative terms, any rearrangement of this series converges to the same (possibly infinite) limit.

Let us look at $S^+$ and $S^-$. As we showed in Proposition 4.6, by definition, we know that $a_n^+ = \max(a_n, 0)$. This implies that $a_n^+ \geq 0$, since if $a_n < 0$, $a_n^+$ returns 0. Thus, we have that all of the terms in $S^+$ are nonnegative. Applying Corollary 3.2, this means any rearrangement of $S^+$ converges to the same limit.

Similarly, as shown before, by definition, we know that $a_n^- = \max(-a_n, 0)$. This similarly implies that $a_n^- \geq 0$, since if $a_n > 0$, $a_n^-$ returns 0. Thus, we have that all the terms in $S^-$ are nonnegative. Again applying Corollary 3.2, this means any rearrangement of $S^-$ converges to the same limit.

Let us take the rearrangement $\sum_{m=1}^{\infty} b_m$. Let $T^+ = \sum_{m=1}^{\infty} b_m^+$ and $T^- = \sum_{m=1}^{\infty} b_m^-$. We observe that $T^+$ is a rearrangement of $S^+$. According to Corollary 3.2, this means $T^+$ converges to the same limit as $S^+$. Since we are given that $S^+$ is finite, we have that $T^+$ is also finite, converging to the same limit as $S^+$. We also observe that $T^-$ is a rearrangement of $S^-$. Similarly, according to Corollary 3.2, this means $T^-$ converges to the same limit as $S^-$. Since we are given that $S^-$ is finite, we have that $T^+$ is also finite, converging to the same limit.

Thus, since we know $T^-$ converges to the same limit as $S^-$ while $T^+$ converges to the same limit as $S^+$, $\sum_{m=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n = S^+ - S^-$.

**Case 2: $S^+$ is not finite, $S^-$ is finite.**

We will use the same method here as the previous case. According to Corollary 3.2, we know that for a series $\sum_{n=1}^{\infty} a_n$ with nonnegative terms, any rearrangement of this series converges to the same (possibly infinite) limit.

Let us look at $S^+$ and $S^-$. As we showed in Proposition 4.6, by definition, we know that $a_n^+ = \max(a_n, 0)$. This implies that $a_n^+ \geq 0$, since if $a_n < 0$, $a_n^+$ returns 0. Thus, we have that all of the terms in $S^+$ are nonnegative. Applying Corollary 3.2, this means any rearrangement of $S^+$ converges to the same limit.

Similarly, as shown before, by definition, we know that $a_n^- = \max(-a_n, 0)$. This similarly implies that $a_n^- \geq 0$, since if $a_n > 0$, $a_n^-$ returns 0. Thus, we have that all the terms in $S^-$ are nonnegative. Again applying Corollary 3.2, this means any rearrangement of $S^-$ converges to the same limit.

Let us take the rearrangement $\sum_{m=1}^{\infty} b_m$. Let $T^+ = \sum_{m=1}^{\infty} b_m^+$ and $T^- = \sum_{m=1}^{\infty} b_m^-$. We observe that $T^+$ is a rearrangement of $S^+$. According to Corollary 3.2, this means $T^+$ converges to the same limit as $S^+$. Since we are given that $S^+$ is not finite, we have that $T^+$ is also not finite. We also observe that $T^-$ is a rearrangement of $S^-$. Similarly, according to Corollary 3.2, this means $T^-$ converges to the same limit as $S^-$. Since we are given that $S^-$ is finite, we have that $T^+$ is also finite, converging to the same limit.

The given note above Corollary 4.7 specifies that if $S^-$ is finite while $S^+$ is infinite, then $\sum_{n=1}^{\infty} a_n = \infty$. Thus, since we know $T^-$ converges to the same limit as $S^-$ while $T^+$ converges to the same limit as $S^+$, $\sum_{m=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n = \infty$.

**Case 3: $S^+$ is finite, $S^-$ is not finite.**

We will use the same method here as the previous cases. According to Corollary 3.2, we know that for a series $\sum_{n=1}^{\infty} a_n$ with nonnegative terms, any rearrangement of this series converges to the same (possibly infinite) limit.

Let us look at $S^+$ and $S^-$. As we showed in Proposition 4.6, by definition, we know that $a_n^+ = \max(a_n, 0)$. This implies that $a_n^+ \geq 0$, since if $a_n < 0$, $a_n^+$ returns 0. Thus, we have that all of the terms in $S^+$ are nonnegative. Applying Corollary 3.2, this means any rearrangement of $S^+$ converges to the same limit.

Similarly, as shown before, by definition, we know that $a_n^- = \max(-a_n, 0)$. This similarly implies that $a_n^- \geq 0$, since if $a_n > 0$, $a_n^-$ returns 0. Thus, we have that all the terms in $S^-$ are nonnegative. Again applying Corollary 3.2, this means any rearrangement of $S^-$ converges to the same limit.

Let us take the rearrangement $\sum_{m=1}^{\infty} b_m$. Let $T^+ = \sum_{m=1}^{\infty} b_m^+$ and $T^- = \sum_{m=1}^{\infty} b_m^-$. We observe that $T^+$ is a rearrangement of $S^+$. According to Corollary 3.2, this means $T^+$ converges to the same limit as $S^+$. Since we are given that $S^+$ is finite, we have that $T^+$ is also finite, converging to the same limit. We also observe that $T^-$ is a rearrangement of $S^-$. Similarly, according to Corollary 3.2, this means $T^-$ converges to the same limit as $S^-$. Since we are given that $S^-$ is not finite, we have that $T^+$ is also not finite.

The given note above Corollary 4.7 specifies that if $S^-$ is infinite while $S^+$ is finite, then $\sum_{n=1}^{\infty} a_n = -\infty$. Thus, since we know $T^-$ converges to the same limit as $S^-$ while $T^+$ converges to the same limit as $S^+$, $\sum_{m=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n = -\infty$.

In all the cases, the limit of the rearrangement matches the limit of the original series. □

**Example 4.8.** This section follows the outline. Let the series $a_n = (-1)^{(n+1)} * \frac{1}{(2^n)}$ for $n \geq 1$. If we expand, we arrive at the following:

$$\sum_{n=1}^{\infty} a_n = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \dots$$

We know that $S^+$ is the sum of the positive terms. This is equal to the following:

$$\begin{aligned} S^+ &= 1 + \frac{1}{4} + \frac{1}{16} + \dots \\ &= \sum_{k=0}^{\infty} \frac{1}{(2^2)^k} \\ &= \frac{1}{1 - \frac{1}{4}} = \frac{4}{3} \end{aligned}$$

We know that $S^-$ is the sum of the magnitude of the negative terms. This is equal to the following:

$$\begin{aligned} S^- &= \frac{1}{2} + \frac{1}{8} + \frac{1}{32} \dots \\ &= \sum_{k=0}^{\infty} \frac{1}{2 * (2^2)^k} \\ &= \frac{\frac{1}{2}}{1 - \frac{1}{4}} \\ &= \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3} \end{aligned}$$

We see here that both $S^+$ and $S^-$ are finite. According to Theorem 4.4, we know then that $\sum_{n=1}^{\infty} a_n = S^+ - S^-$. Thus, we have that $\sum_{n=1}^{\infty} a_n = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$. It is also worth noting

that since we see that $S^+$ and $S^-$ are finite, according to Corollary 4.7 we have that any rearrangement of this series will still converge to $\frac{2}{3}$.

Note that so far we have considered cases where either $S^+$ or $S^-$ is finite. The situation will be different if both $S^+$ and $S^-$ are infinite.

**Theorem 4.9.** *Let $\sum_{n=1}^{\infty} a_n$ be a series such that $S^+$ and $S^-$ are both infinite, and also $\lim_{n\to\infty} a_n = 0$. Given any number $r \in \mathbb{R}$, there is a rearrangement of the series that converges to $r$.*

*Proof.* This section follows the outline and Abbott 2.9 closely. Let us take $\sum_{k=1}^{\infty} b_k$, a rearrangement of the series $\sum_{n=1}^{\infty} a_n$.
Begin with $S_0 = 0$, our initial sum. Then, our constructive algorithm will go as follows:
Add positive terms from $a_n^+$ until our sum is greater than $r$. Next, add negative terms from $a_n^-$ until our sum is less than $r$. Now, again, add positive terms from $a_n^+$ until our sum is greater than $r$. When it is greater than $r$, return to adding negative terms. Repeat this to construct the rearrangement. Since $\lim_{n\to\infty} a_n = 0$, as the rearrangement goes on, its partial sum oscillates closer and closer around $r$ and thus converges to $r$. Formally, since we have infinitely many terms in $a_n^+$ and $a_n^-$, alternating them in a way where the partial sum oscillates closer and closer to $r$ guarantees that for any $\epsilon > 0$, there are sufficiently small positive and negative terms to add so that the oscillation (or distance between $S_n$ and $r$) is less than $\epsilon$. This results in convergence to $r$. □

**Example 4.10.** This section follows the outline. Let us take the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.
We see that the positive terms can be described by $a_{2k-1}$, since only the odd-indexed terms in $a_n$ raise $(-1)$ to an even power and thus produce a positive term. Thus, we can express the sum of the positive terms as follows:

$$S^+ = \sum_{k=1}^{\infty} \frac{1}{2k-1}$$

since by plugging in $(2k-1)$ for $n$ in $a_n = \frac{(-1)^{n-1}}{n}$, we get $\frac{1}{2k-1}$ for $k = 1, 2, \ldots$. We see that $\frac{1}{2k-1} > \frac{1}{k}$, so by the comparison test, we know that $\frac{1}{2k-1}$ must diverge to $\infty$ (we know the harmonic series diverges to $\infty$). Thus, we have that $S^+$ diverges to $\infty$.

We see that the negative terms can be described by $a_{2k}$, since only the even-indexed terms in $a_n$ raise $(-1)$ to an odd power and thus produce a negative term. Thus, we can express

the sum of the positive terms as follows:

$$S^- = \sum_{k=1}^{\infty} \frac{1}{2k}$$

since by plugging in $(2k)$ for $n$ in $a_n = \dfrac{(-1)^{n-1}}{n}$, we get $\dfrac{1}{2k}$ for $k = 1, 2, \ldots$. Again, by the comparison test, we see that $\dfrac{1}{2k} > \frac{1}{k}$, which implies that $\dfrac{1}{2k}$ must diverge to $\infty$ since we know the harmonic series diverges to $\infty$. In other words, $S^-$ will diverge to $\infty$.

Let us choose $r = \frac{2}{9}$. As we described in Theorem 4.9, we should be able to use that algorithm to rearrange the terms so that our series converges to $r = \frac{2}{9}$.

We begin with $S_0 = 0$. We add our first positive term $a_1 = 1$. $S_1 = 1$.
We subtract our first negative term, $a_2 = \frac{-1}{2}$. $S_2 = 1 - \frac{1}{2} = \frac{1}{2}$.
We have not yet fallen below $r$, so we continue to subtract negative terms. We subtract $a_4 = \frac{-1}{4}$, so $S_3 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.
We have not yet fallen below $r$, so we continue to subtract negative terms. We subtract $a_6 = \frac{-1}{6}$, so $S_4 = \frac{1}{4} - \frac{1}{6} \approx .0833$.
We have now fallen below $r$, so let us add our next positive term, $a_3 = \frac{1}{3}$. $S_5 = .0833 + \frac{1}{3} \approx .4167$.
We have now exceeded $r$, so let us return to subtracting negative terms. We can do follow the algorithm laid out in Theorem 4.9 in this way, until we have the first 15 terms as follows:

$$\sum_{n=1}^{\infty} b_n = 1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{6} + \frac{1}{3} - \frac{1}{8} - \frac{1}{10} + \frac{1}{5} - \frac{1}{12} - \frac{1}{14} - \frac{1}{16} + \frac{1}{7} - \frac{1}{18} - \frac{1}{20} + \frac{1}{9} - \ldots$$

We continue this algorithm infinitely and can observe that it converges to $r = \frac{2}{9}$. Briefly, we can observe that it converges to $\frac{2}{9}$ by seeing that the oscillations around $\frac{2}{9}$ are getting closer and closer to $\frac{2}{9}$ as $|a_n| \to 0$. Thus, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N, |S_n - \frac{2}{9}| < \epsilon$ - in other words, since $|a_n| \to 0$, we are guaranteed that there is always an oscillation smaller than $\epsilon$.